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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	All analyses in the paper used previously collected data. For details about data collection, see the relevant prior work: Isometric force task: Ma X*, Rizzoglio F*, Bodkin KL, Perreault EJ, Miller LE, Kennedy A. eLife 2:e84296. DOI: https://doi.org/10.7554/eLife.84296 Reaching task: Perich MG, Gallego JA, Miller LE (2018) A neural population mechanism for rapid learning. Neuron, 100 (4), 771-773. https://doi.org/10.1016/j.neuron.2018.09.030 Gallego JA*, Perich MG*, Chowdhury RH, Solla SA#, Miller LE# (2020) Long-term stability of cortical population dynamics underlying consistent behavior. Nature Neuroscience, 23, 260–270. https://doi.org/10.1038/s41593-019-0555-4 Ma X*, Rizzoglio F*, Perreault EJ, Miller LE, Kennedy A. bioRxiv 2022.08.26.504777; doi: https://doi.org/10.1101/2022.08.26.504777 Human iBCI data: Rubin DB, Hosman T, Kelemen JN, Kapitonava A, Willett FR, Coughlin BF, Halgren E, Kimchi EY, Williams ZM, Simeral JD, Hochberg LR, Cash SS (2022) Learned Motor Patterns Are Replayed in Human Motor Cortex during Sleep. Journal of Neuroscience 42(25):5007-5020. doi: 10.1523/JNEUROSCI.2074-21.2022
Data analysis	Custom analysis and plotting scripts were written using Python (3.7.7). The NoMAD model was implemented in Python (3.7.7) with code made available at https://github.com/snel-repo/nomad. The core LFADS model was implemented using TensorFlow (2.0.0) in Python (3.7.7) and is available at https://snel-repo.github.io/autolfads. Aligned FA was implemented in MATLAB using the previously published public code base

available at https://github.com/alandegenhart/stabilizedbci. ADAN was implemented in Python (3.7.7) using the previously published public code base available at https://github.com/farshchian/ADAN.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Monkey data has been made previously available at: https://doi.org/10.5061/dryad.cvdncjt7n
Human data can be found at: https://doi.org/10.5061/dryad.q83bk3jtp

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	This study was performed on two male rhesus macaques and one male human clinical trial participant. Due to the use of previously collected data and clinical trial recruitment criteria (for the human participant), we did not consider sex or gender in this study.
Population characteristics	Participant T11 is a 36-year-old right-handed man with a history of tetraplegia secondary to a cervical spinal cord injury. He had two 96-channel intracortical microelectrode arrays placed chronically into the left precentral gyrus (PCG) as part of the BrainGate pilot clinical trial (www.ClinicalTrials.gov; Identifier: NCT00912041), with placement occurring 153 days and 427 days prior to the recording sessions.
Recruitment	The participant met recruitment criteria outlined as part of the clinical trial detailed above. The participant gave informed consent to the study and resulting publications.
Ethics oversight	Permission for this study was granted by the U.S. FDA (Investigational Device Exemption #G090003) and the IRBs of Massachusetts General Hospital, Providence VA Medical Center, and Brown University.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We used a sample size = 3 (two monkeys + one human/three tasks). In addition, each experiment (i.e., each subject/task) consists of 12 sessions spanning 38 days, 20 sessions spanning 95 days, and 2 sessions spanning <= 2 hours each, with each session containing 200-300 trials each. This amount of sessions and trials is sufficient to reliably uncover differences between methods. These sample sizes are consistent with previous demonstrations of new computational methodologies applied to stabilizing decoding performance over time.
Data exclusions	We excluded trials during which the subject failed to complete the task and some rewarded trials with abnormal behavior for the sake of visualization. All data exclusion criteria are delineated in Methods.
Replication	We performed the same analyses for two monkeys and two different tasks (isometric force and reaching). The results hold for both monkeys and both tasks. Code will be made available upon publication.
Randomization	Validation data for training of all models is selected at random. In addition, testing data for reporting the performance of decoding performance is selected at random.
Blinding	All data were originally collected for other purposes, so data collection was not influenced by the goals of this study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Rhesus macaque, adult male
Wild animals	N/A
Reporting on sex	We used previously collected data originally collected for other scientific studies and therefore were limited in the individuals that were available. We did not report results on sex as this is not a relevant factor for our study.
Field-collected samples	N/A
Ethics oversight	All surgical and experimental procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of Northwestern University under protocol #IS00000367 and are consistent with the Guide for the Care and Use of Laboratory Animals.

Note that full information on the approval of the study protocol must also be provided in the manuscript.